Vernier: calibration-aware acquisition design for IVIM diffusion MRI

Do b-value schemes differ in post-conformal uncertainty calibration at matched scan-time and matched Cramér–Rao precision?

(author list)

Draft — feasibility build

Scope of evidence. The feasibility result (§4.1) is *publication-independent*: it runs entirely on the open, synthetic, MIT-licensed Caliper toolkit and depends on no paper in review. Claims marked P are *provisional*: the efficiency-frontier decision values (§4.2) use the decision lens of **Minos**, and the honest-scope statement uses the identifiability-wall result of **Gauge**; both companions are under review. If either changes in revision, the P numbers are invalidated and must be re-run (`../ASSUMPTIONS.md`). Every number is regenerated from seeded results by `consistency.py`.

Abstract

Diffusion-MRI acquisition design optimises either the *variance* of the point estimate (the Cramér–Rao bound, CRLB) or the *expected information* about the parameters (Bayesian experimental design, BED / EIG). Neither asks whether the resulting *uncertainty* is *trustworthy*: whether, after a distribution-free conformal correction, the reported error bars are well *calibrated*. We pose and test that question for intravoxel incoherent motion (IVIM) imaging, where the pseudo-diffusion coefficient D^* is notoriously ill-determined. The question is nontrivial because split-conformal restores *marginal* coverage to nominal for every acquisition by construction; what it does *not* equalise—conditional coverage and interval sharpness—is the test. On matched-scan-time, matched-CRLB(D^*) b-value schemes that differ only in how they spend a fixed budget, the post-conformal D^* interval width spans a factor of $393/289$ (width-spread $\Delta_{\text{sharp}} = 0.328$, 95% CI $[0.200, 0.399]$) and the high-D^* conditional coverage spans 0.842–0.901 ($\Delta_{\text{cond}} = 0.059$, CI $[0.036, 0.098]$), even though marginal coverage is held at ≈ 0.908 for all of them. Both gaps clear pre-registered thresholds with bootstrap CIs excluding zero: **acquisition design moves the part of calibration conformal does not fix, at matched precision.** The effect is *estimator-contingent*: it is large for the over-confident segmented estimator that conformal correction targets but *collapses* below threshold for an efficient amortised (MAF) posterior, which is already near-calibrated ($\Delta_{\text{sharp}} : 0.328 \to 0.041$, $\Delta_{\text{cond}} : 0.059 \to 0.030$); the divergence is a property of the estimator×acquisition interaction, and matters precisely for the imperfect estimators that dominate clinical IVIM. We are deliberately narrow on a second axis too: Vernier does *not* claim to improve identifiability—a companion analysis P shows the high-D^* wall is acquisition-robust (CRLB-optimal design moved CRLB(D^*)/tercile-width only $1.25 \to 1.05$). The claim lives on the calibration-and-decision axis.

1 Introduction

The choice of diffusion weightings (b-values) in an IVIM acquisition is an experiment-design problem. Two objectives dominate the literature. The first is *variance-optimal* design: place b-values to minimise the CRLB of the fitted parameters [1, 2, 3]. The second is *information-gain* design: maximise the expected information gain (EIG) about the parameters under a Bayesian experimental-design (BED) framework [4, 5, 6]. Both optimise a property of the *point estimate*—its variance, or the information the data carry about it.

Neither asks the question a clinician actually relies on: when the method reports an error bar, can you believe it, and how much is that error bar *worth* per minute of scan time? Trustworthiness of an interval is *calibration*: does a nominal 90% interval cover the truth 90% of the time, marginally and within clinically relevant strata? Distribution-free conformal prediction [8, 9, 7] restores marginal coverage for essentially any base estimator. This raises a sharp question that, to our knowledge, the acquisition-design literature has not posed:

> *At matched scan-time and matched CRLB precision, do different b-value schemes yield differently-* ***calibrated*** *uncertainty* ***after*** *conformal correction—and therefore different decision value per scan-minute?*

This is not answered by either canon. Marginal coverage is equalised by conformal correction by construction, so it cannot distinguish schemes; and the CRLB / EIG objectives never look at calibration at all. We test the question directly. Because the answer was genuinely unknown to us in advance—and because a negative answer would mean Vernier is not a standalone contribution but a section of the decision-calibration companion—we treated it as a pre-registered *feasibility gate* (§4.1), with thresholds fixed before running and no tuning toward a positive outcome.

2 Honest scope: calibration, not identifiability

We must separate two claims that are easy to conflate. *Identifiability* asks whether the data determine the parameter at all; for IVIM's D^* the answer is largely no, and—crucially—this does not improve with acquisition design. A companion conformal-coverage analysis[P] swept clinical, CRLB-optimal, and dense b-schemes and found the high-D^* identifiability wall *acquisition-robust*: high-D^* coverage barely moved ($0.841 \rightarrow 0.844$) and the ratio CRLB(D^*)/tercile-width stayed $\geq$ 1.05 for every scheme, so CRLB-optimal design lowered the bound ($1.25 \rightarrow 1.05$) without removing the wall. We take that wall as given.

Vernier therefore makes *no* identifiability claim. It asks the orthogonal question of whether acquisition design moves *calibration*—the trustworthiness and sharpness of the error bar after correction—and the downstream decision value. Our own CRLB implementation (§3) reproduces the companion's ≈ 1.05–1.25 ratio as a cross-check, but the feasibility result below does not depend on it.

3 Methods

Cohort and estimator (synthetic, open). We generate synthetic IVIM cohorts with the open Caliper toolkit: true parameters (D, f, D^*) drawn from physiologically plausible priors (D^* log-uniform on $[10, 100] \times 10^{-3}$ mm^2/s, the poorly identified axis), clean bi-exponential signals on a chosen b-schedule, and Rician noise at a chosen SNR. The device under test is Caliper's torch-free *segmented* least-squares estimator, which reports deliberately over-confident, homoscedastic quantiles—the standard model-based-UQ failure mode conformal is meant to repair. No clinical data are used.

Matched scan-time and matched precision. Scan-time is proportional to the number of acquired volumes; we hold it fixed by using 11 b-values for every scheme in the gate. Among such schemes we select a subset whose expected CRLB(D^*)—computed from the IVIM Fisher information matrix—agree to within $\pm 10\%$ of the candidate-pool median, so neither budget nor variance can explain a difference between them. Matching precision is conservative: it removes precision as a confounder and makes any surviving calibration divergence *harder* to find, not easier.

Conformal correction and the ruler. For each scheme we split the cohort into calibration and test halves, fit Caliper's split-conformal / conformalised-quantile-regression (CQR) correction [7] on the calibration half, and apply it to the test half. We score the corrected intervals with a model-agnostic calibration ruler: marginal coverage,

scheme	CRLB(D^*)	raw cov.	post cov.	width	high-D^* cov.
perfusion-lean	20.5	0.325	0.904	300	0.865
tissue-lean	23.2	0.294	0.895	330	0.842
wide-ends	22.2	0.295	0.908	289	0.901
mid-heavy	23.4	0.252	0.903	393	0.849

Table 1: Feasibility gate at SNR 33. All four schemes share scan-time (11 b-values) and CRLB(D^*) (within $\pm 10\%$). Conformal equalises *marginal* coverage (≈ 0.90); the post-conformal *width* and *high-D^* conditional coverage* do not equalise. Numbers trace to `results/feasibility_gate.json`.

quantile expected calibration error (ECE), sharpness (mean interval width), and conditional coverage within true-D^* terciles [10, 11].

Pre-registered gate. The decision was fixed before running. Marginal coverage is *not* the metric: CQR restores it to nominal for every scheme by construction (we report it only as a sanity check). The load-bearing quantities are what conformal does not equalise: *sharpness*, summarised by the across-scheme width spread $\Delta_{\text{sharp}} = (\max - \min)/\text{median}$ of the post-conformal D^* interval width; and *conditional coverage*, summarised by the range Δ_{cond} of high-D^*-tercile coverage. We declare the question **feasible** (PASS) iff $\Delta_{\text{sharp}} \geq 0.10$ or $\Delta_{\text{cond}} \geq 0.05$, *and* that gap's 95% bootstrap CI excludes zero. The bootstrap is *paired*: schemes share the same true parameters and noise (only the b-scheme differs), and each resample draws both the calibration indices (which set each scheme's conformal offset, and hence its width) and the test indices (which set coverage).

4 Results

4.1 The feasibility gate: schemes diverge in post-conformal calibration

Table 1 reports 4 matched-scan-time, matched-CRLB(D^*) schemes (CRLB(D^*) $\in [20.5, 23.4]$), at SNR 33, over 8000 voxels with a 2000-iteration paired bootstrap. CQR restores the marginal D^* coverage of every scheme from a raw 0.252–0.325 to 0.895–0.908 ($\approx$ nominal 0.90). Yet the post-conformal D^* interval *width* ranges from 289 to 393 ($\Delta_{\text{sharp}} = 0.328$, 95% CI [0.200, 0.399]), and the *high-D^* conditional coverage* ranges from 0.842 to 0.901 ($\Delta_{\text{cond}} = 0.059$, CI [0.036, 0.098]). Both gaps clear their pre-registered thresholds with CIs excluding zero: the verdict is **PASS**.

The effect is robust to SNR. At SNR 25, $\Delta_{\text{sharp}} = 0.311$ and $\Delta_{\text{cond}} = 0.056$; at SNR 50, $\Delta_{\text{sharp}} = 0.038$ (the width gap shrinks as the estimator's D^* error converges) while $\Delta_{\text{cond}} = 0.072$ persists. The conditional-coverage divergence is thus the robust load-bearing finding: marginal conformal correction hides scheme-dependent conditional miscoverage that the acquisition determines.

4.2 Efficiency frontier: decision value saturates with scan-time

We next score, through the decision lens of the companion decision-calibration work [P] [12], each scheme's conformal-corrected D^* posterior on a stylised treat/spare/escalate decision with asymmetric cost. Sweeping scan-time from 7 to 22 b-values, the corrected D^* interval sharpens ($316 \rightarrow 198$) and the achieved decision utility improves, but with steeply *diminishing returns*: the marginal decision utility per added scan-minute falls $9.9 \rightarrow 2.9 \rightarrow 0.47$ across the steps. Crucially, *no* protocol beats the no-scan prior baseline for D^* (every mean utility lies below the prior's -14.6)[P]: acquisition does not rescue D^* *decisions*, consistent with the identifiability wall (§2). The protocol implication is that the clinical 11-b acquisition captures most of the recoverable calibrated decision value; beyond it, additional scan-minutes buy very little, and none of it closes the D^* decision gap.

4.3 Robustness: the divergence is estimator-contingent

The gate above used Caliper's deliberately over-confident segmented estimator. To test whether the divergence is a property of the *acquisition* or of the *estimator*, we re-ran the identical gate—same matched schemes, train/cal/test splits, metrics, and paired bootstrap—on Caliper's efficient masked-autoregressive-flow (MAF) posterior, and on the reference estimator under those same splits. (Both estimators ship with the open Caliper toolkit, so this experiment is also SOLID.) Under matched splits the reference still PASSES ($\Delta_{\text{sharp}} = 0.331$, $\Delta_{\text{cond}} = 0.079$). The MAF, by contrast, is already near-calibrated *before* any correction (raw D^* coverage ≈ 0.88); its corrected D^* intervals are an order of magnitude sharper (≈ 46–48 vs 289–393); and, crucially, they barely differ across schemes: $\Delta_{\text{sharp}} = 0.041$ (CI $[0.016, 0.072]$) and $\Delta_{\text{cond}} = 0.030$ (CI $[0.013, 0.067]$), both below the pre-registered thresholds: **FAIL**. The calibration divergence is therefore not a property of the acquisition alone but of the estimator$\times$acquisition interaction: it appears when the estimator's realised error is large and varies with sampling geometry—the over-confident regime conformal correction exists to repair—and vanishes for an efficient posterior whose error already tracks the CRLB.

5 Discussion

The feasibility result establishes a third axis for diffusion-MRI acquisition design, distinct from variance (CRLB) and information (BED/EIG): *post-correction calibration*—but, sharpened by §4.3, only for the over-confident estimators that conformal correction targets. At matched precision and matched scan-time, where the variance- and information-optimal objectives are by definition indifferent, schemes built on such an estimator still differ substantially in the sharpness and conditional coverage of their conformally corrected D^* intervals. Because marginal conformal correction equalises only marginal coverage, these differences are invisible to a marginal-coverage audit and to the CRLB, yet they are exactly what a downstream decision sees.

We are deliberately careful about what this does and does not show. **(1)** It is a calibration claim, not an identifiability claim: the companion wall result[P] stands, and our own frontier confirms that acquisition does not make D^* decisions beat the prior. **(2)** The divergence is *estimator-contingent*, not universal (§4.3): it is large for the over-confident segmented estimator but collapses below threshold for an efficient MAF posterior. We read this as localising the contribution rather than refuting it—acquisition-aware calibration matters exactly for the imperfect, over-confident estimators that dominate clinical IVIM and that motivate conformal correction in the first place, and is unnecessary when a well-calibrated amortised posterior is already available. **(3)** The sharpness gap is SNR-dependent (it shrinks at high SNR), whereas the conditional-coverage gap persists, so the latter is the more durable signal. **(4)** "CI excludes zero" is a weak bar for a non-negative range statistic; the substantive criterion is the pre-registered *magnitude* threshold, which both gaps clear.

6 Limitations and reproducibility

All data are synthetic and PHI-free. The feasibility gate (§4.1) and the estimator-contingency result (§4.3) are SOLID: both run on the open Caliper toolkit alone (the segmented and MAF estimators ship with it) and depend on no manuscript in review. The decision-value frontier (§4.2) and the honest-scope wall statement are PROVISIONAL—they consume the in-review decision-calibration[P] and conformal-coverage[P] companions—and are re-validated by one command when those land (`../ASSUMPTIONS.md`, `../PROMOTION.md`). Every number above is emitted by `consistency.py` from the seeded result files (`results/feasibility_gate.json`, `results/efficiency_frontier.json`, `results/maf_gate.json`); the manuscript contains no hand-entered figures.

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